GNU Emacs Cheatsheet

Complete Developer Reference * GNU Emacs 31 * ~/.emacs.d/init.el

GNU Emacs 31.1 (x86_64)
Go * Zig * C/C++ * Magit

KEY NOTATION: `C- = Ctrl` `M- = Alt (Meta)` `S- = Shift` `RET = Enter` `SPC = Space` *LSP: Eglot * Completion: Vertico / Corfu*

1. CORE MOVEMENT & CURSOR NAVIGATION

`movement`

KEY / SHORTCUT	COMMAND & ACTION	CONTEXT / NVIM EQV
`C-f / C-b`	Move forward / backward by character	*forward/backward-char (NVim: l / h)*
`C-n / C-p`	Move downward / upward by line	*next/previous-line (NVim: j / k)*
`M-f / M-b`	Move forward / backward by word	*forward/backward-word (NVim: w / b)*
`C-a / C-e`	Move to beginning / end of current line	*beginning/end-of-line (NVim: 0 / $)*
`M-a / M-e`	Move to beginning / end of sentence	*backward/forward-sentence*
`M-{ / M-}`	Move to previous / next paragraph / code block	*backward/forward-paragraph (NVim: { / })*
`M-< / M->`	Move to beginning / end of entire buffer	*beginning/end-of-buffer (NVim: gg / G)*
`C-v / M-v`	Scroll page down / page up	*scroll-up/down-command (NVim: <C-f>/<C-b>)*
`C-l`	Recenter screen on cursor (cycles: center -> top -> bottom)	*recenter-top-bottom (NVim: zz)*
`M-g g`	Go to specific line number (e.g. M-g g 42 RET)	*goto-line (NVim: :{line} / G)*

2. CORE EDITING, DELETION & CLIPBOARD

`editing`

KEY / SHORTCUT	COMMAND & ACTION	CONTEXT / NVIM EQV
`C-d / DEL`	Delete character forward / backward	*delete-char / backward-delete (NVim: x / X)*
`M-d / M-DEL`	Delete word forward / backward	*kill-word / backward-kill-word (NVim: dw / db)*
`C-k`	Kill (cut) from cursor position to end of line	*kill-line (NVim: D)*
`C-y`	Yank (paste) most recently killed text	*yank (NVim: p)*
`M-y`	Cycle through kill ring history (or consult-yank-pop)	*yank-pop*
`C-/ or C-_`	Undo last change	*undo (NVim: u)*
`C-SPC`	Set selection mark (start highlighting region)	*set-mark-command (NVim: v)*
`C-x h`	Mark / select entire buffer	*mark-whole-buffer (NVim: ggVG)*
`TAB`	Smart indent line or trigger completion candidate	*indent-for-tab-command*
`C-g`	Cancel / abort active command or minibuffer prompt	*keyboard-quit (NVim: <Esc>)*

3. CUSTOM EDITING ERGONOMICS

`matklad & tsoding`

KEY / SHORTCUT	COMMAND & ACTION	CONTEXT / NVIM EQV
`C-,`	Duplicate line downward keeping cursor column	*rc/duplicate-line (NVim: yyp)*
`C-w`	Smart cut: cuts selection, or whole line if none	*kill-region-smart (Matklad / Tsoding)*
`M-w`	Smart copy: copies selection, or whole line if none	*kill-ring-save-smart*
`M-p / M-n`	Slide current line or selected block UP / DOWN	*move-text (Tsoding)*
`C-=`	Expand selection outward: word -> quote -> expr -> block	*expand-region*
`C-S-f`	AST folding: toggle fold on leaf functions & tests (Zig)	*my/fold-functions (Matklad)*
`S-<arrow>`	Switch focus directly between split windows	*windmove*

4. PROJECT MANAGEMENT & SEARCH

`project.el & consult`

KEY / SHORTCUT	COMMAND & ACTION	CONTEXT / NVIM EQV
`C-x p f`	Find file in project (interactive fuzzy matcher)	*project.el (NVim: <leader>ff)*
`C-x p p`	Switch active project workspace	*project.el*
`C-x p b`	Switch buffer belonging to current project	*project.el (NVim: <leader>fb)*
`C-x p c`	Run compile command at project root directory	*project-compile*
`C-x p d`	Open project root directory in Dired file manager	*project-dired*
`C-x p e`	Open Eshell terminal inside project root	*project-eshell*
`C-c f g`	Project-wide live ripgrep with live file preview	*consult-ripgrep (NVim: <leader>fg)*
`C-x b`	Switch buffer, recent file, or bookmark with preview	*consult-buffer*

5. LSP & CODE INTELLIGENCE

`eglot & tree-sitter`

KEY / SHORTCUT	COMMAND & ACTION	CONTEXT / NVIM EQV
`M-.`	Go to definition of symbol under point	*xref-find-definitions (NVim: gd)*
`M-?`	Find all references across project	*xref-find-references (NVim: gr)*
`M-,`	Jump backward to origin after definition jump	*xref-go-back (NVim: <C-t>)*
`M-[ / M-]`	Jump backward / forward in navigation jump history	*better-jumper (NVim: <C-o>/<C-i>)*
`C-c r`	Rename symbol project-wide	*eglot-rename (NVim: <leader>rn)*
`C-c a`	Code actions & quickfixes at point	*eglot-code-actions (NVim: <leader>ca)*
`C-c h`	Show documentation / signature hover popup	*eldoc (NVim: K)*
`C-c C-f`	Format buffer using LSP server	*eglot-format-buffer (gofmt / zls)*
`M-g i`	Search functions, methods, and types in current file	*consult-imenu (NVim: <leader>fs)*
`M-g I`	Search symbols across all project buffers	*consult-imenu-multi (NVim: <leader>fS)*

6. BUILDING, RUNNING & DEBUGGING

`compile, flymake, gud`

KEY / SHORTCUT	COMMAND & ACTION	CONTEXT / NVIM EQV
`M-x compile`	Run build command (ANSI colored, no save prompts)	*compile (Matklad tuning)*
`M-x recompile`	Re-run last compile command instantly without prompting	*recompile*
`M-g n / M-g p`	Jump to NEXT / PREVIOUS compiler error or grep hit	*next-error / prev-error*
`RET`	(in compile buffer) Jump directly to source line of error	*compile-goto-error*
`g`	(in compile buffer) Re-run compilation in place	*recompile*
`M-g d`	Show Flymake diagnostics list for current file	*flymake (NVim: <leader>d)*
`M-x gdb`	Launch GDB interactive multi-window debug session	*GUD debugger (C / Zig / Go)*
`C-x C-a C-b`	Toggle breakpoint on current line	*gud-break*
`C-x C-a C-n`	Step over next statement	*gud-next*
`C-x C-a C-s`	Step into function	*gud-step*
`C-x C-a C-c`	Continue execution	*gud-cont*
`C-x C-a C-r`	Print value of expression under point	*gud-print*
`M-x eshell / term`	Built-in Elisp/shell hybrid or full terminal emulator	*terminal*

7. SYNTAX NAVIGATION & STRUCTURAL EDITING

`sexp & rectangle`

KEY / SHORTCUT	COMMAND & ACTION	CONTEXT / NVIM EQV
`M-;`	Comment or uncomment line / region; or add line comment	*comment-dwim (NVim: gcc / gc)*
`C-M-\`	Re-indent selected region according to language rules	*indent-region (NVim: =)*
`C-M-a / C-M-e`	Jump to beginning / end of current function	*beginning/end-of-defun*
`C-M-f / C-M-b`	Jump forward / backward over balanced expression / block	*forward/backward-sexp*
`C-M-u / C-M-d`	Move up / down one level of block nesting	*backward/down-list*
`C-M-k`	Kill (cut) entire balanced expression / syntax block forward	*kill-sexp*
`C-M-t`	Transpose / swap two adjacent expressions or arguments	*transpose-sexps*
`C-M-SPC`	Mark (select) entire balanced syntax expression / block	*mark-sexp*
`C-x SPC`	Rectangle mark mode: column block selection	*rectangle-mark (NVim: <C-v>)*
`C-x r t`	Replace column rectangle with typed string across lines	*string-rectangle*
`C-c @ C-c`	Hide / toggle code block at cursor	*hs-hide-block (prog-mode)*
`C-c @ C-a`	Show all code blocks across current file	*hs-show-all*

8. FULL GIT WORKFLOW

`magit (C-x g)`

KEY / SHORTCUT	COMMAND & ACTION	CONTEXT / NVIM EQV
`C-x g`	Open Magit status buffer (full Git dashboard)	*magit-status (NVim: <leader>gg)*
`s / u`	Stage / unstage file or hunk under point	*magit-stage / unstage*
`v (visual) + s`	Stage ONLY highlighted lines within a hunk	*partial hunk staging*
`k`	Discard hunk or file changes with confirmation	*magit-discard (NVim: X)*
`S / U`	Stage ALL files / unstage ALL files	*magit stage-all*
`c c`	Commit staged changes (C-c C-c saves, C-c C-k aborts)	*magit-commit*
`c a`	Amend previous commit	*magit-commit-amend*
`P p / F p`	Push to remote repository / pull from remote	*magit-push / pull*
`b b / b c`	Checkout branch / create and checkout new branch	*magit-branch*
`z z / z p`	Stash changes / pop latest stash	*magit-stash*
`r i`	Interactive rebase from commit under point	*magit-rebase*
`l l / d d`	Show Git commit log graph / show working diff	*magit-log / diff*
`M-x magit-blame`	Show inline Git blame annotations per line (q exits)	*magit-blame (NVim: gitsigns)*
`C-c M-g`	Magit file dispatch menu (file log, blame, diff)	*magit-file-dispatch*

9. WINDOWS, SPLITS & LAYOUTS

`window & winner`

KEY / SHORTCUT	COMMAND & ACTION	CONTEXT / NVIM EQV
`C-x 2 / C-x 3`	Split window horizontally (top/bottom) / vertically (side-by-side)	*split-window*
`C-x 0 / C-x 1`	Close current window / close ALL other windows (maximize)	*delete-window(s)*
`C-x o`	Cycle cursor focus to next split window	*other-window*
`Shift + Arrow`	Switch focus directly to window in arrow direction	*windmove*
`C-x 4 f`	Open file in OTHER split window	*find-file-other-window*
`C-x 4 b`	Switch buffer in OTHER split window	*switch-to-buffer-other-window*
`C-c <left>`	Undo window layout change (restore accidental close/resize)	*winner-undo*
`C-x 5 2 / C-x 5 0`	Create new GUI window frame / close current frame	*make/delete-frame*
`C-x +`	Balance / equalize window sizes across split panes	*balance-windows*

10. BUFFER & FILE OPERATIONS

`files & buffers`

KEY / SHORTCUT	COMMAND & ACTION	CONTEXT / NVIM EQV
`C-x C-f`	Find and open file (or project-find-file C-x p f)	*find-file*
`C-x C-s`	Save current buffer to disk	*save-buffer*
`C-x s`	Save all modified buffers (prompts per buffer)	*save-some-buffers*
`C-x b`	Switch buffer / recent files / bookmarks	*consult-buffer*
`C-x k`	Kill / close current buffer	*kill-current-buffer (NVim: <leader>q)*
`C-x C-q`	Toggle buffer read-only state	*read-only-mode*
`C-x C-c`	Exit GNU Emacs	*save-buffers-kill-terminal*

11. DIRED FILE MANAGER

`dired & wdired`

KEY / SHORTCUT	COMMAND & ACTION	CONTEXT / NVIM EQV
`C-x d`	Open Dired directory browser	*dired*
`C-x C-q (in Dired)`	WDired: make file list editable text to batch-rename files!	*wdired-mode*
`+`	Create new directory	*dired-create-directory*
`C / R`	Copy file / Rename or move file (auto-targets 2nd split pane)	*dired-do-copy / rename*
`d / u`	Flag file for deletion / unflag file	*dired-flag-file-deletion*
`x`	Execute deletion of all flagged files	*dired-do-flagged-delete*
`g`	Refresh directory file listing	*revert-buffer*

12. TREEMACS FILE EXPLORER

`treemacs`

KEY / SHORTCUT	COMMAND & ACTION	CONTEXT / NVIM EQV
`C-c t t`	Toggle Treemacs sidebar file explorer	*treemacs (NVim: <leader>e)*
`C-c t f`	Locate and highlight current file in Treemacs tree	*treemacs-find-file*
`C-c t p`	Add project directory to Treemacs workspace	*treemacs-add-project*
`RET or o`	(in Treemacs) Open file / expand or collapse directory	*treemacs-RET-action*
`c f / c d`	(in Treemacs) Create new file / directory under point	*treemacs-create-file/dir*
`d / R`	(in Treemacs) Delete file / rename file at point	*treemacs-delete / rename*
`g r`	(in Treemacs) Refresh project file tree	*treemacs-refresh*
`t h`	(in Treemacs) Toggle display of hidden dotfiles	*treemacs-toggle-show-dotfiles*
`q / ?`	(in Treemacs) Close Treemacs window / show popup help	*treemacs-quit / help*

13. SEARCH, REPLACE & MINIBUFFER

`vertico, corfu, orderless`

KEY / SHORTCUT	COMMAND & ACTION	CONTEXT / NVIM EQV
`C-s / C-r`	Live line search in current buffer with preview	*consult-line (isearch)*
`M-%`	Interactive query-replace (y=yes, n=skip, !=all remaining)	*query-replace*
`C-M-%`	Regex interactive query-replace	*query-replace-regexp*
`M-y`	Interactive visual kill-ring history picker	*consult-yank-pop*
`SPC (in search)`	Orderless filter: match search tokens in any order	*orderless*
`C-n / C-p`	Next / previous candidate in Vertico list or Corfu popup	*vertico / corfu*
`M-n / M-p`	History next / previous element in Minibuffer prompt	*minibuffer-history*
`TAB / RET`	Accept candidate in Corfu / select candidate in Vertico	*corfu-insert / vertico-directory*
`M-RET`	Submit input verbatim without auto-selecting match	*vertico-exit-input*
`C-g`	Cancel / escape current command or minibuffer prompt	*keyboard-quit (NVim: <Esc>)*

14. DISCOVERY & ENVIRONMENT

`help & themes`

KEY / SHORTCUT	COMMAND & ACTION	CONTEXT / NVIM EQV
`C-h k + <key>`	Describe key: displays exact function bound to any key	*describe-key*
`C-h f + <func>`	Describe function: full documentation and source code	*describe-function*
`C-h v + <var>`	Describe variable: docstring, current value, customization	*describe-variable*
`C-h m`	Describe mode: lists all active major/minor modes and keymaps	*describe-mode*
`M-x load-theme`	Switch color themes (naysayer, gruber-darker, gruvbox, solarized)	*load-theme*
`M-x treesit-install...`	Download and compile missing Tree-sitter language grammar	*treesit-install-language-grammar*

TOOLCHAIN & LARGE CODEBASES

* GC tuning: gc-cons-threshold = 100MB prevents typing freezes.
* JSON-RPC pipes: read-process-output-max = 1MB for gopls / zls.
* TigerBeetle: auto-selects ZLS 0.14 if ./zig/zig exists in project root.
* Tree-sitter: grammars stored in ~/.emacs.d/tree-sitter/ (*.so).

KEY HABITS: EMACS VS NEOVIM

* Escape key: In Emacs, use C-g (keyboard-quit) to cancel anything.
* Leader key: C-x p is your <leader> for project operations.
* Magit is modal: Inside Magit (C-x g), single keys (s, u, c, P) work like Vim.
* WDired: C-x C-q in file listings turns filenames into editable text.